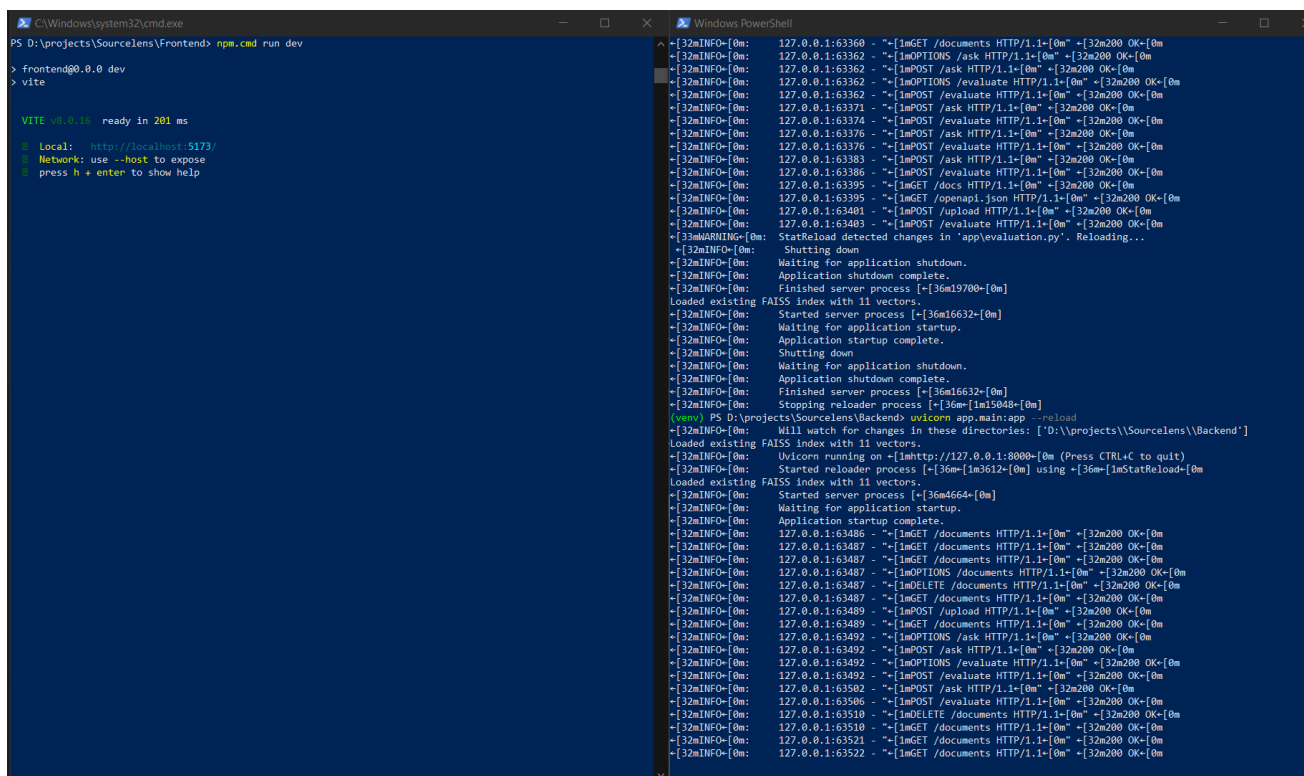


SourceLens: RAG Document Q&A

System Walkthrough

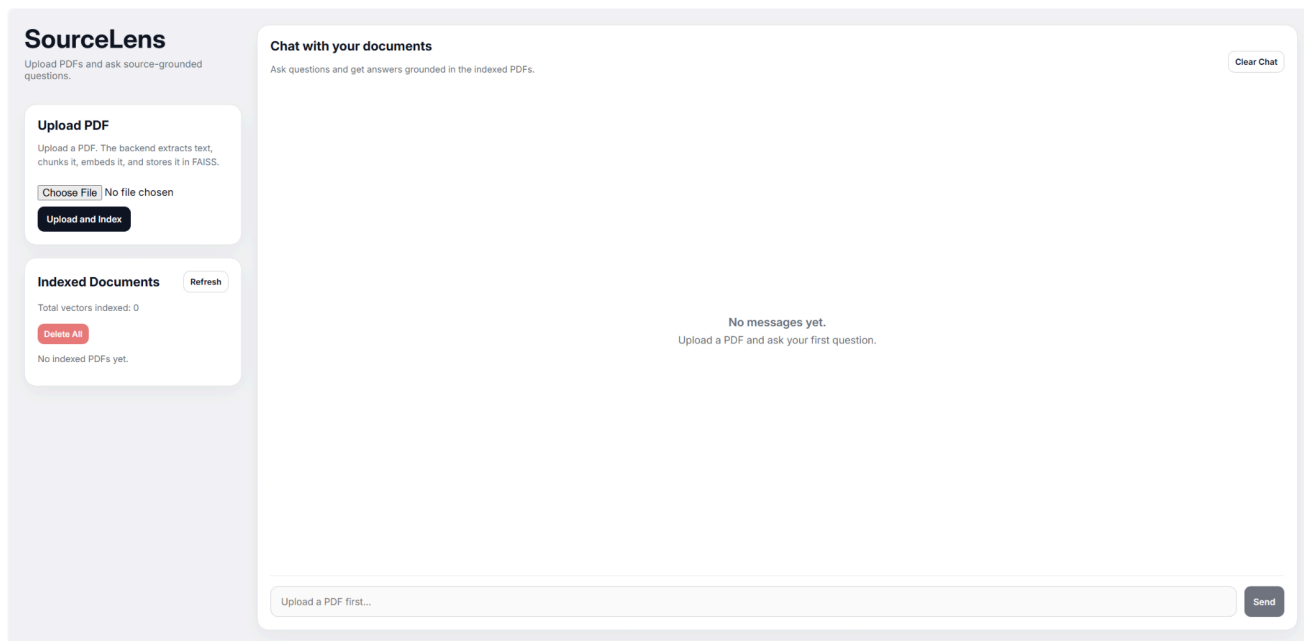
This screenshot below shows the SourceLens system running locally. The left terminal starts the frontend using `npm run dev`, while the right terminal shows the backend server running and processing requests.



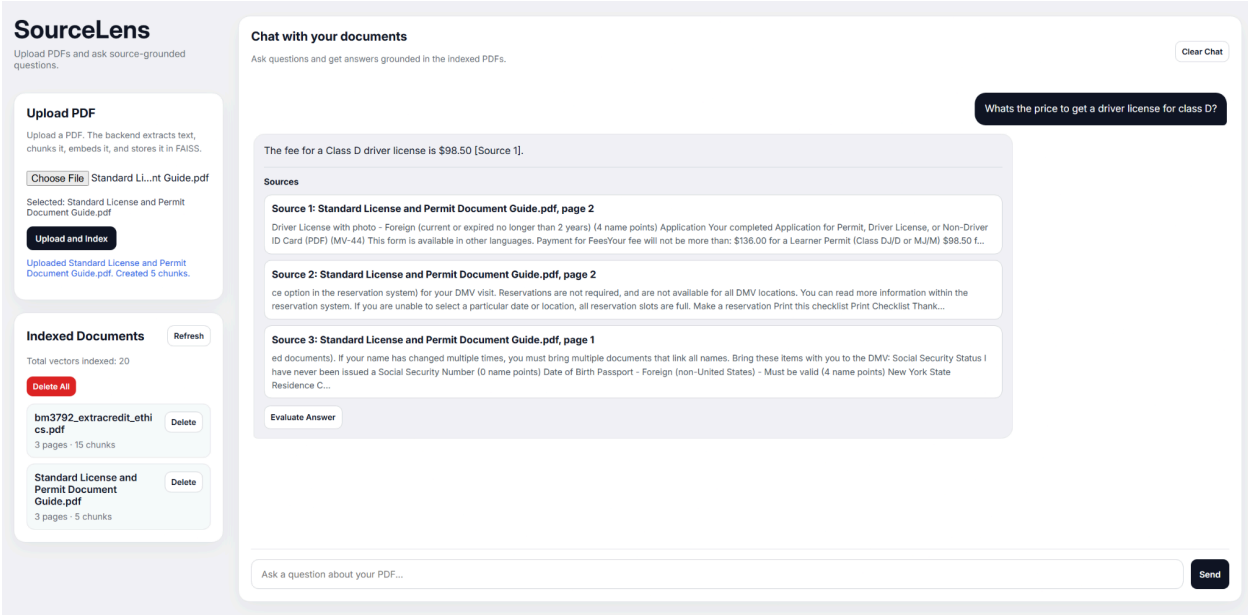
```
C:\Windows\system32\cmd.exe
PS D:\projects\SourceLens\Frontend> npm.cmd run dev
> frontend@0.0.0 dev
> vite

VITE v4.0.0 ready in 201 ms
  Local:   http://localhost:5173/
  Network: use --host to expose
  press h + enter to show help

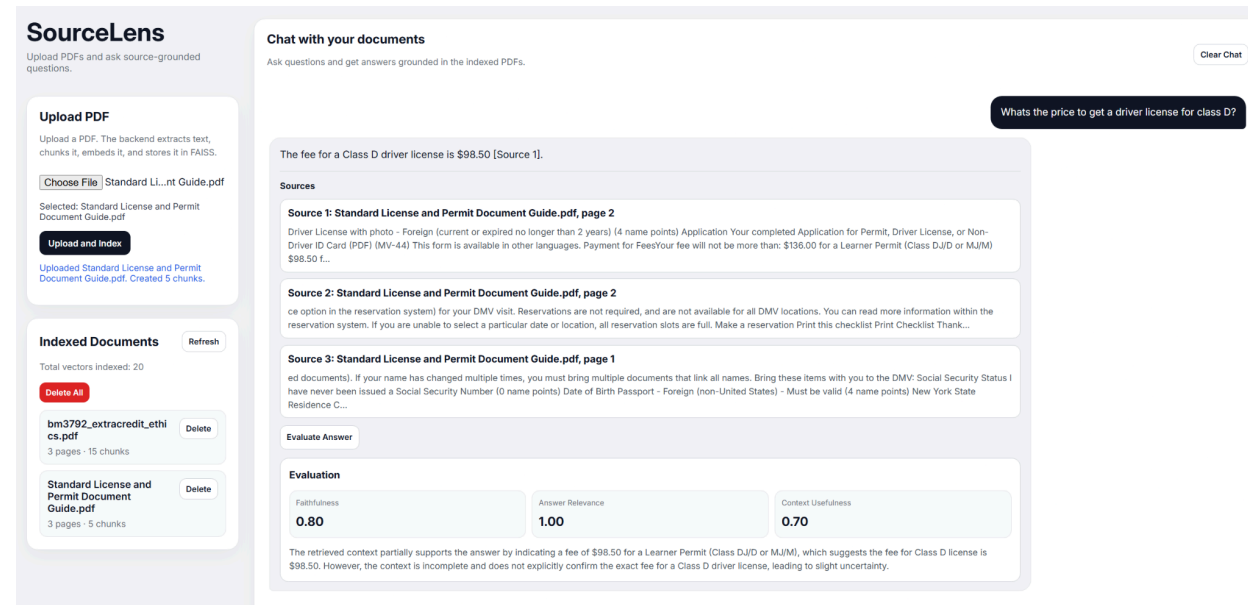
C:\Windows\system32\cmd.exe
PS D:\projects\SourceLens\Backend> uvicorn app:main --reload
INFO: 127.0.0.1:63360 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63362 - "[1mOPTIONS /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63362 - "[1mPOST /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63362 - "[1mOPTIONS /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63362 - "[1mPOST /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63371 - "[1mPOST /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63374 - "[1mPOST /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63376 - "[1mPOST /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63376 - "[1mPOST /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63383 - "[1mPOST /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63386 - "[1mPOST /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63395 - "[1mGET /docs HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63395 - "[1mGET /openapi.json HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63401 - "[1mPOST /upload HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63403 - "[1mPOST /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
WARNING: StatReload detected changes in 'app/evaluation.py'. Reloading...
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [-[36m19700-[0m]
INFO: Loaded existing FAISS index with 11 vectors.
INFO: Started server process [-[36m16632-[0m]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [-[36m16632-[0m]
INFO: Stopping reloader process [-[36m[1m5040-[0m]
INFO: PS D:\projects\SourceLens\Backend> uvicorn app:main --reload
INFO: Will watch for changes in these directories: ['D:\projects\SourceLens\Backend']
INFO: Loaded existing FAISS index with 11 vectors.
INFO: uvicorn running on +[1mhttp://127.0.0.1:8000-[0m (Press CTRL+C to quit)
INFO: Started reloader process [-[36m[1m3612-[0m] using [-[36m[1mStatReload-[0m]
INFO: Loaded existing FAISS index with 11 vectors.
INFO: Started server process [-[36m4664-[0m]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:63486 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63487 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63487 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63487 - "[1mOPTIONS /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63487 - "[1mDELETE /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63487 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63489 - "[1mPOST /upload HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63489 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63492 - "[1mOPTIONS /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63492 - "[1mPOST /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63492 - "[1mOPTIONS /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63492 - "[1mPOST /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63502 - "[1mPOST /ask HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63506 - "[1mPOST /evaluate HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63510 - "[1mDELETE /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63510 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63521 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
INFO: 127.0.0.1:63522 - "[1mGET /documents HTTP/1.1-[0m" +[32m200 OK-[0m
```



The above screenshot shows the main SourceLens user interface before asking any questions. The left sidebar contains the PDF upload section, where users can choose a file and upload it for indexing. It also includes the “Indexed Documents” section, which lists uploaded documents and their chunk counts. The main panel is the chat area where users can ask questions after a document has been indexed.



The above screenshot shows SourceLens after a PDF has been uploaded and indexed. The user asks a question about getting a driver license for class D. SourceLens generates an answer using the uploaded PDF and displays the supporting sources below the response. Each source includes the document name, page number, and matching text, helping users verify where the answer came from.



The last screenshot shows the same question-answer workflow with an added evaluation section. After generating the answer and showing the retrieved sources, SourceLens displays evaluation scores such as faithfulness, answer relevance, and context precision. These scores help measure how well the generated answer is supported by the uploaded document and how relevant it is to the user's question.